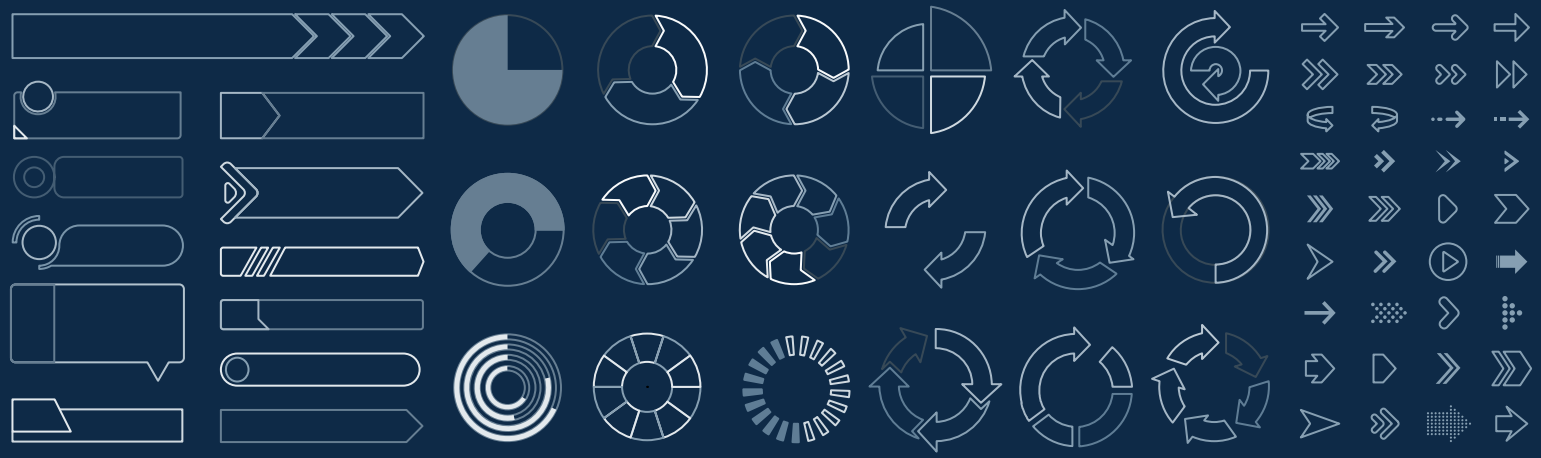
AI in Software Testing

By Kapila Karunaratne

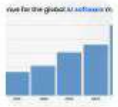
Head of Testing | Delivery Head | Director - QA | Researcher & Lecturer

The New Age of Software Testing

Emerging Trends !



What are the New Trends in Software Engineering?



Artificial intelligence

Agile - DevOps

Cloud computing

Cybersecurity

DevSecOps

DevOps

Microservices

Kubernetes

Blockchain

Low Code No Code

Quantum computing

IoT

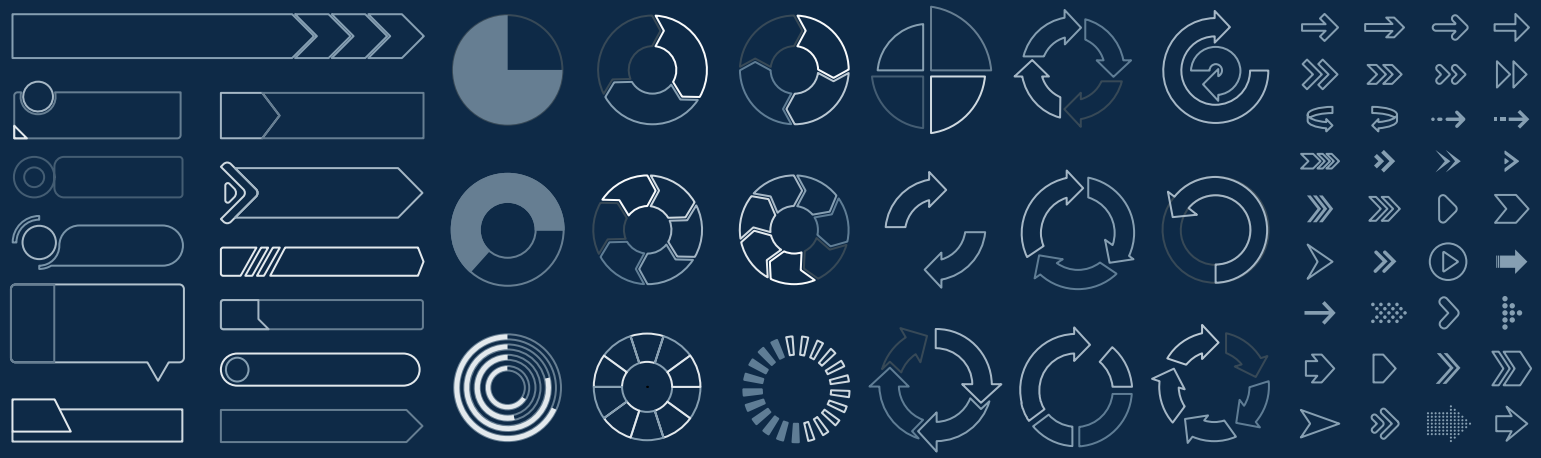
Do not edit
How to change the design

What are the TOP 5 influencing New Trends in Software Engineering for you?

Presenting with animations, GIFs or speaker notes? Enable our Chrome extension

slido

Why Artificial Intelligence



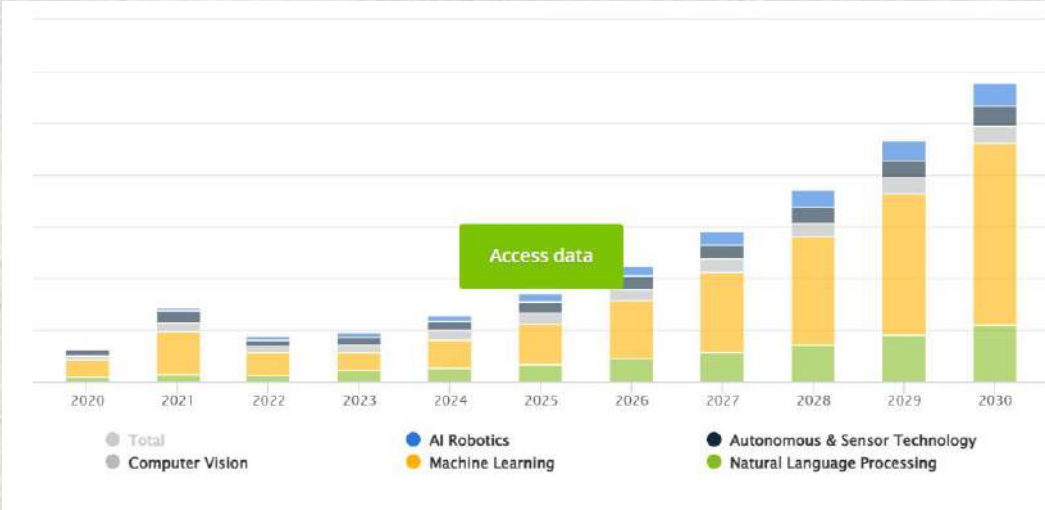
Artificial Intelligence in 2030

- **$15.7** Trillion Potential contribution to the global economy by 2030 from AI
- Up to **26%** boost in GDP for local economies from AI by 2030
- Approx. **300** AI use cases identified and rated are captured in AI Impact Index

Source: PwC analysis

Generative AI - Worldwide

- The market size in the Generative AI market is projected to reach **$36.06bn** in **2024**
- The market size is expected to show an annual growth rate (CAGR 2024-2030) of **46.47%**, resulting in a market volume of **$356.10bn** by 2030



Source: STATISTA

AI impact on Industry in 2030

20m

Number of manufacturing jobs that could be displaced by industrial robots by 2030—8.5% of the global manufacturing workforce.

Source: OXFORD ECONOMICS

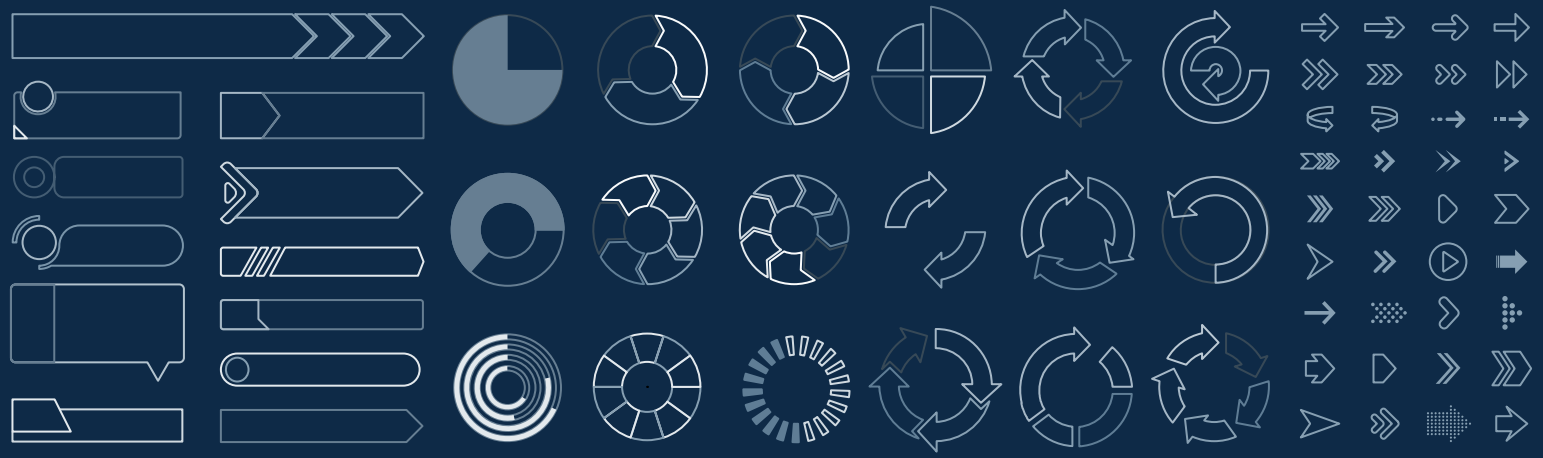
AI impact on IT Industry in 2030

“India’s AI journey is at a defining moment with agentic AI reshaping the workforce and creating over 3 million new tech jobs by 2030,”

“The shift is set to dramatically impact sectors like manufacturing (8M jobs), retail (7.6M), and education (2.5M). While roles like payroll clerks are being phased out, positions such as AI Configurators (66% hiring rate), Data Scientists (65%), and Experience Designers (57%) are on the rise, “

- Sumeet Mathur, SVP and MD, ServiceNow India Technology and Business Centre.

Emerging Trends in Software Testing !



What are the new trends?

Test Automation

Agile, TDD/BDD and DevOps

Software Testing in Cybersecurity

Artificial Intelligence for Testing

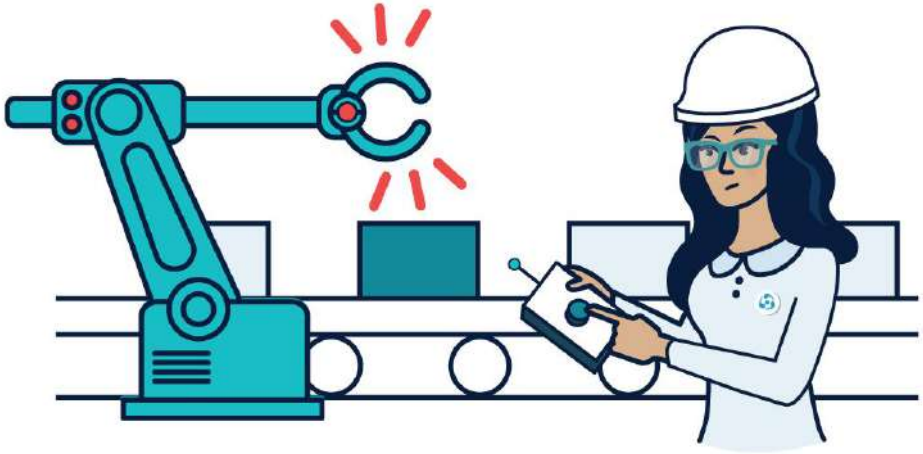
Data Science in Testing

Blockchain Testing

Test Automation

Test automation will

- speed up the release cycles
- increase coverage
- ensures quality software releases



API and Services Test Automation

Decoupling the client and server is a current trend in designing both Web and mobile applications.

tosca ACCELQ th2 Accello DTA

Scriptless Automation (No / Low Code)

These tests will be built using

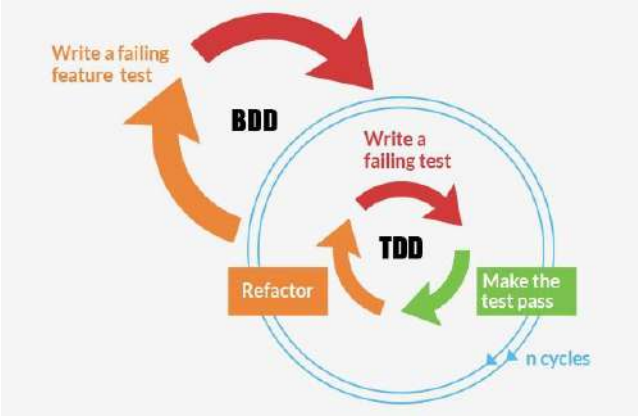
- Model
- Keyword
- object
- data-based approaches

Scriptless test automation enables testers to rapidly create automated tests around the functionalities of the software application with little coding knowledge.

Agile , TDD/BDD and DevOps

- Agile and DevOps bring together lean thinking and automation, aligning collaboration between development and operations.
- The goal of Agile and DevOps is quality and value optimization

TDD/BDD



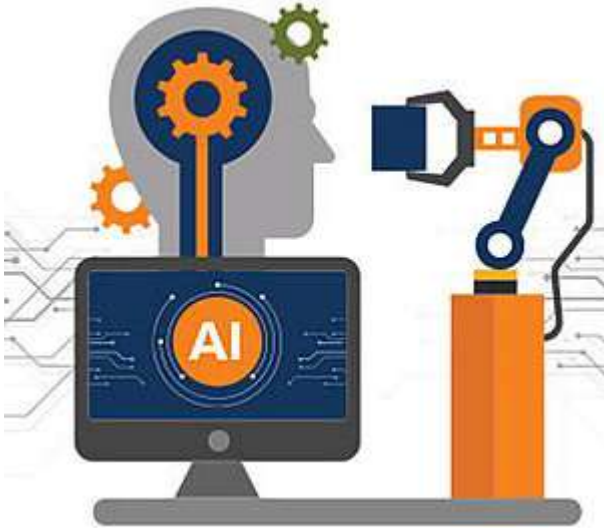
DevOps

- DevOps involves practices, rules, processes, and tools that help to integrate development and operation activities to reduce the time from development to delivery to operations
- The adoption of both Agile and DevOps helps the teams to develop and deliver quality software faster
- Test automation as it is an essential element of the DevOps process.

Artificial Intelligence / Data Science for Testing

AI in Software Testing

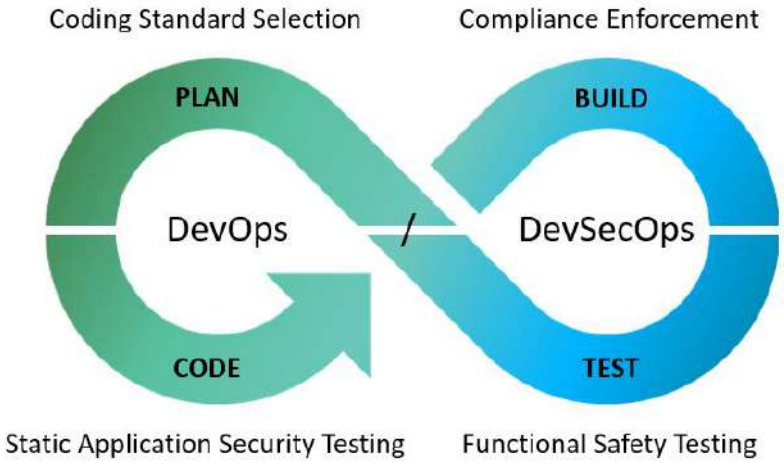
- Software testing is usually a time-consuming process
- AI is the best choice for faster deployments with insufficient infrastructure
- Artificial Intelligence-powered automation can minimize testing work to human ingenious and cognitive ability
- It would ensure more fail-safe results and susceptible to imprecisions and discrepancies
- AI helps in speeding up the software testing process as the repetitive tasks can be efficiently performed



Benefits of AI in Software Testing

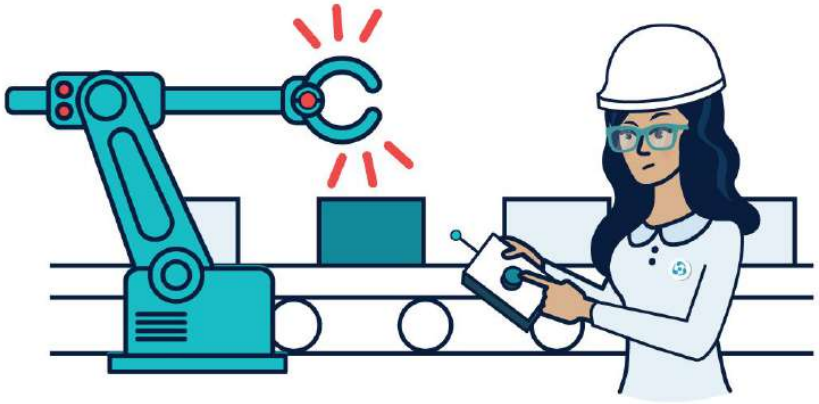
- Reduces the test-cycle time
- Helps to find bugs faster
- Encourages Script Execution
- Improves the speed of Test Coverage
- Reduces Costs Requirements
- Improves Accuracy

Software Testing in Cybersecurity



- Software testing streamlines the cybersecurity factor restricting hacking and diminishing the chances of any threat to data
- The adoption of security testing services will aid the security process against unexpected cyber-attacks to industries like healthcare, finance, banking, etc. while improving the entire development design
- Software testing could help drive a mature security practice that will run through different stages of the development lifecycle while aligning with cybersecurity objectives

AI for Test Automation



Automated Visual Validation

Verify that the UI appears correctly to users with use of AI/ML

API and Services Test Automation

Decoupling the client and server is a current trend in designing both Web and mobile applications.

More automated tests

Using ML it could decide what the smallest number of tests is to test the piece of changed code and automatically write tests

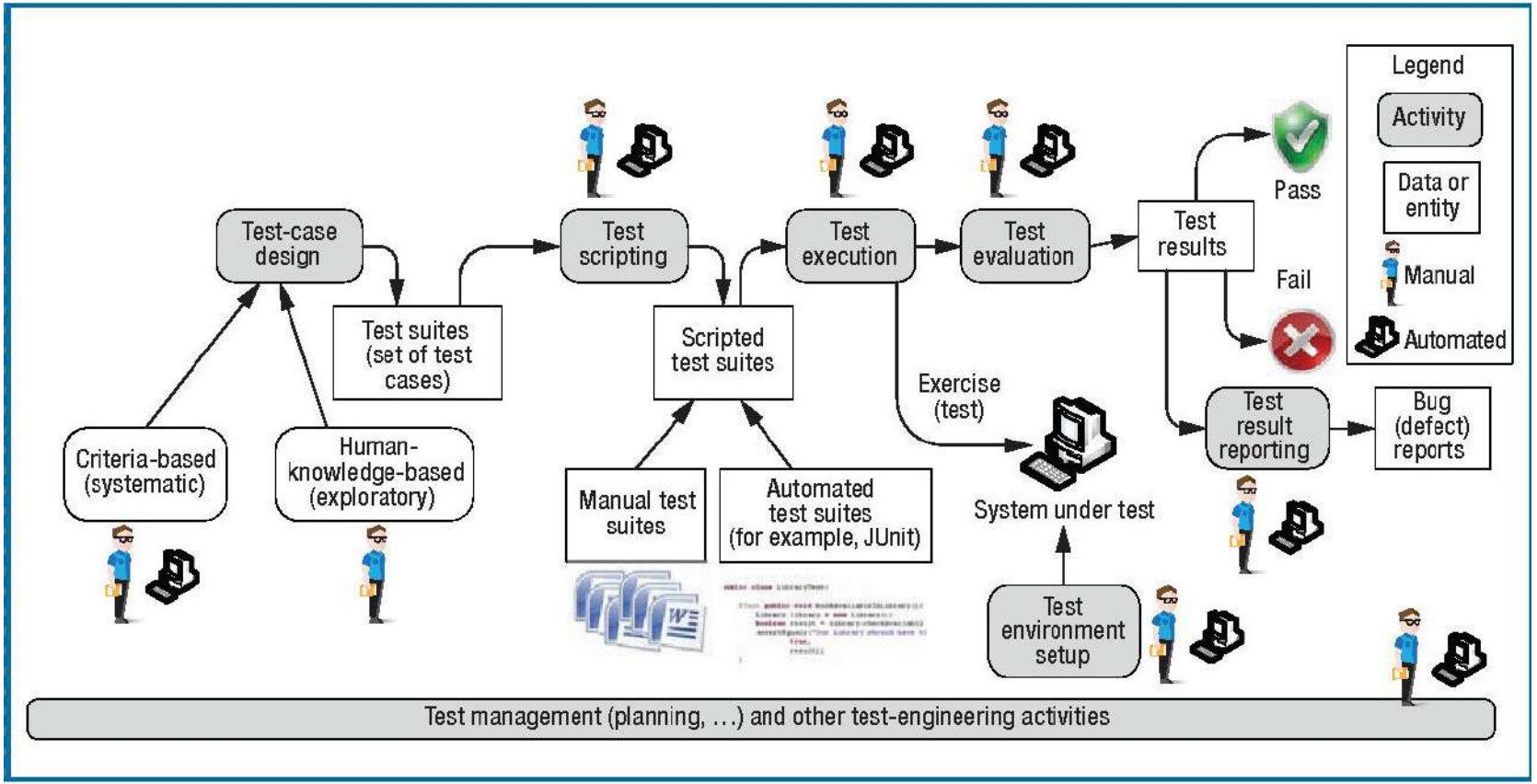
More reliable automated tests

Use AI and machine learning to automatically adjust the scripts

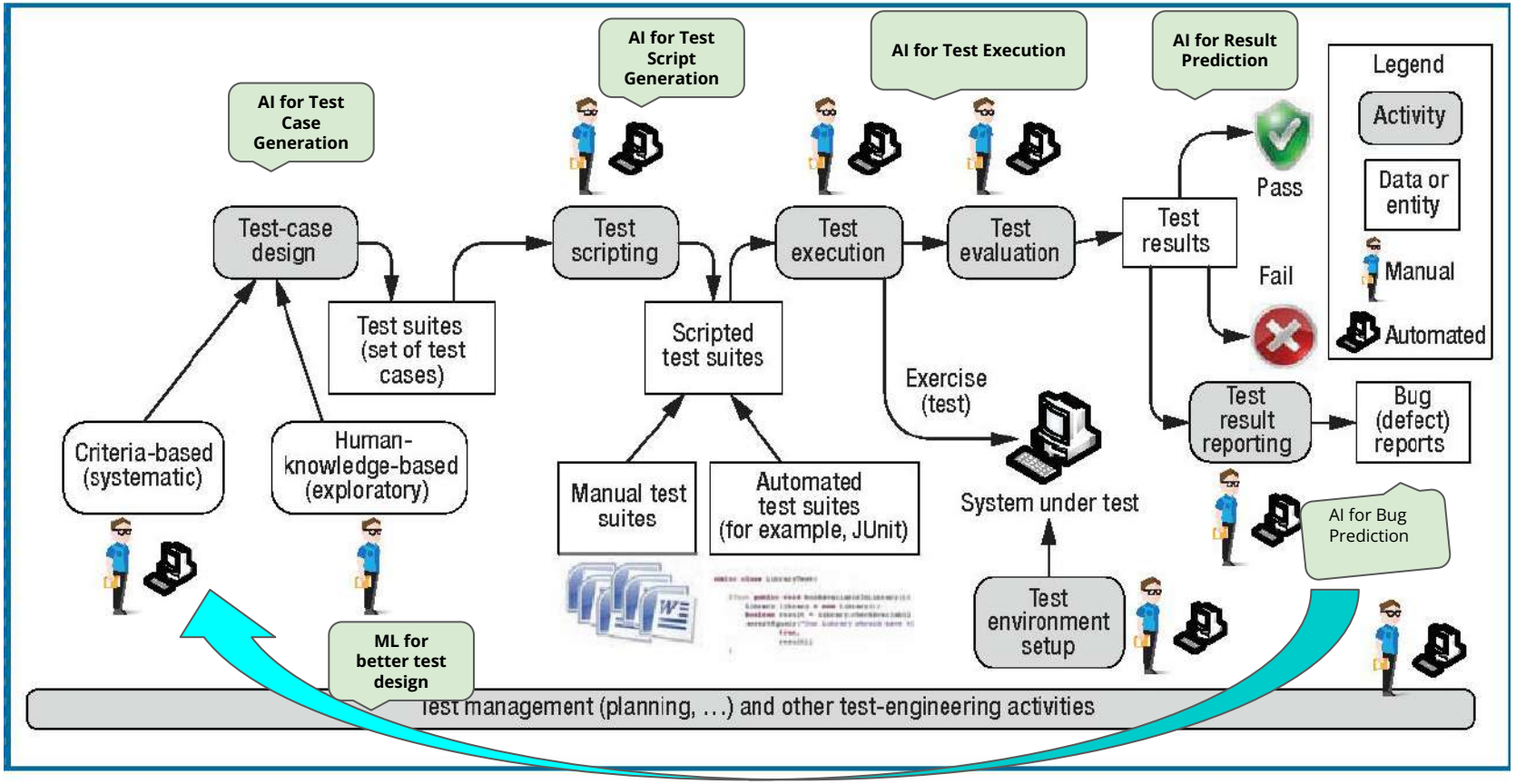
AI and ML in Test Automation

Leveraging Artificial Intelligence and machine learning to enhance and optimize **test automation, including test case generation, test script maintenance , test execution and result prediction** etc.

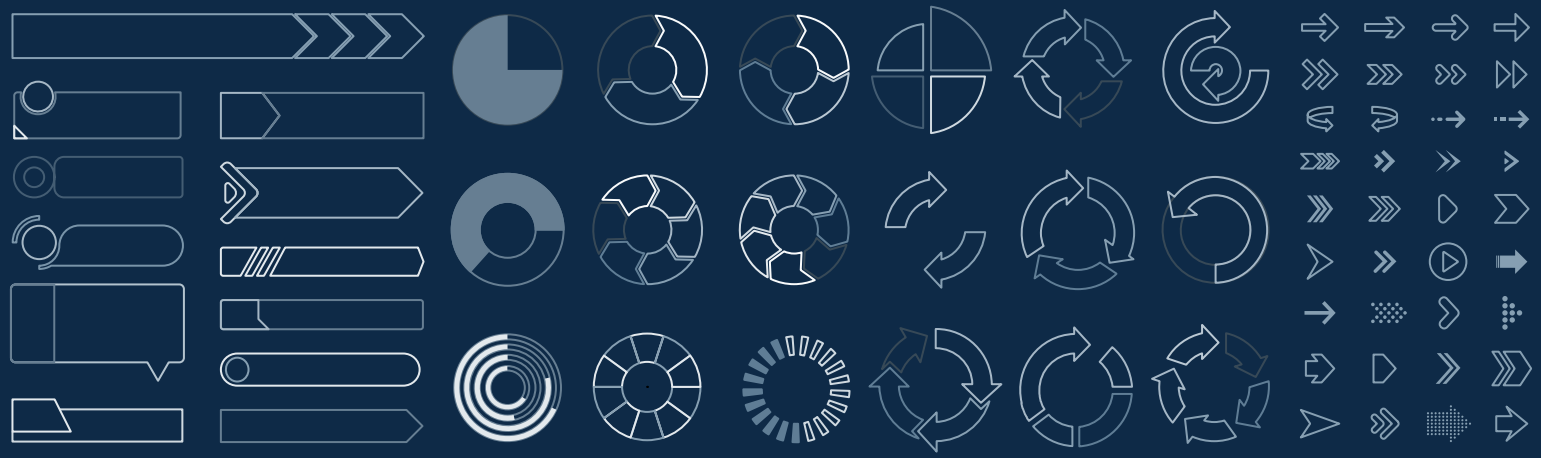
End to End Automation in actual test project



How to use AI/AL in Automation projects



Evolution of AI



Evolution of Artificial Intelligence

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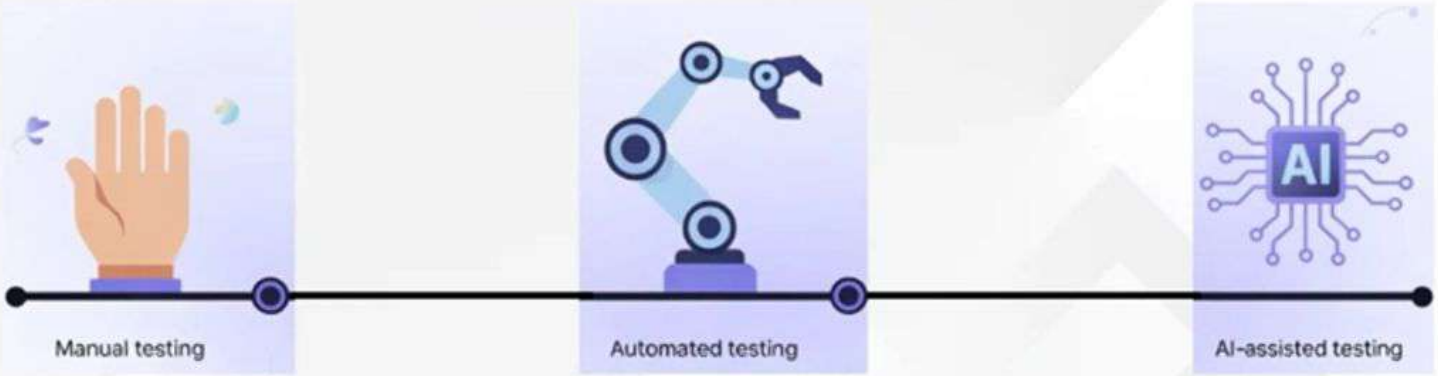


Past:

Human-centered, intuitive but slow and inconsistent.

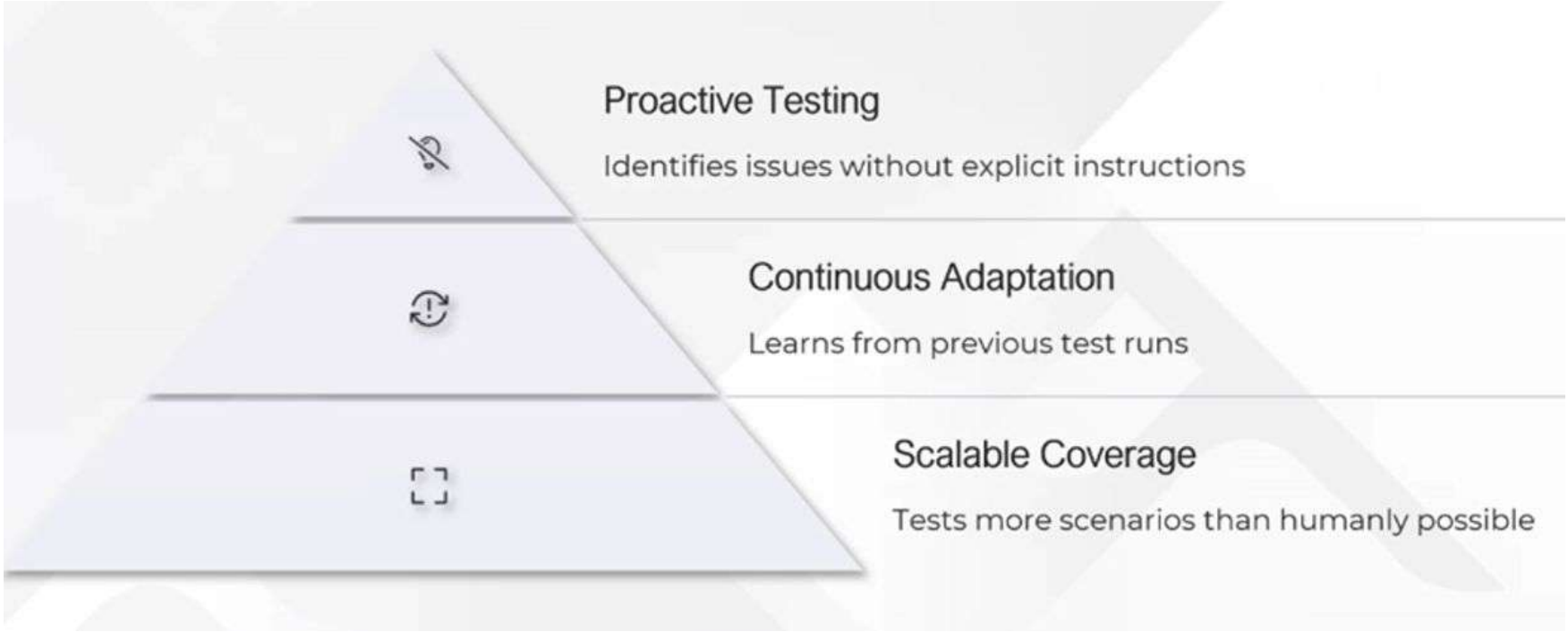
Present:

Script-based, faster but maintenance-heavy.

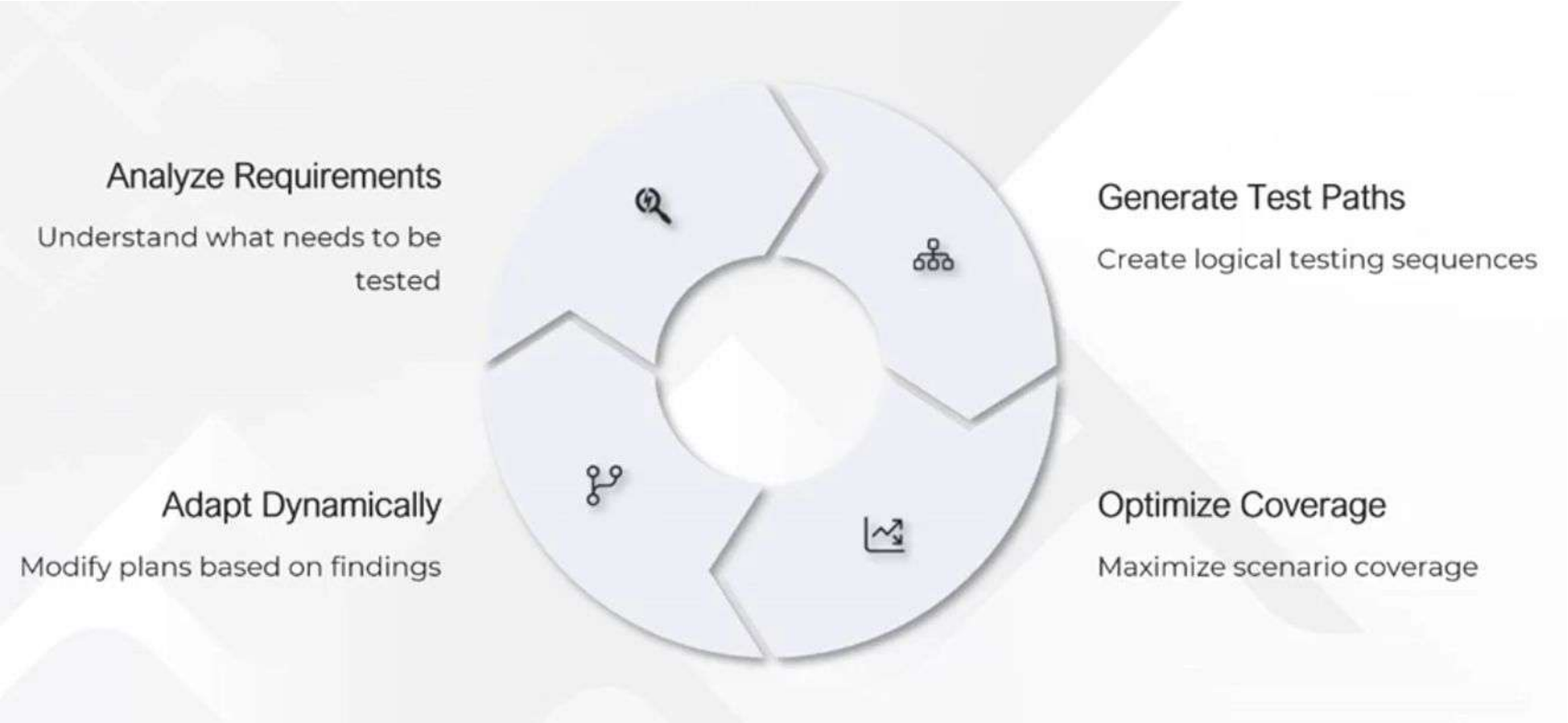
Future: Agentic Testing

Goal-oriented, adaptive, and collaborative with humans.

How Agentic AI matters for QA



What Agentic AI can do for QA



Key Opportunities with Agentic AI for QA

10x

Scalability

Increased test coverage without proportional resource increase.

24/7

Continuous Testing

Non-stop quality monitoring across time zones.

90%

Automation Rate

Higher percentage of automated test scenarios.

New Skill Set for QA

Effective Prompting

Guiding AI behavior through clear instructions and examples.

Critical Assessment

Evaluating AI-generated test results with human judgment.

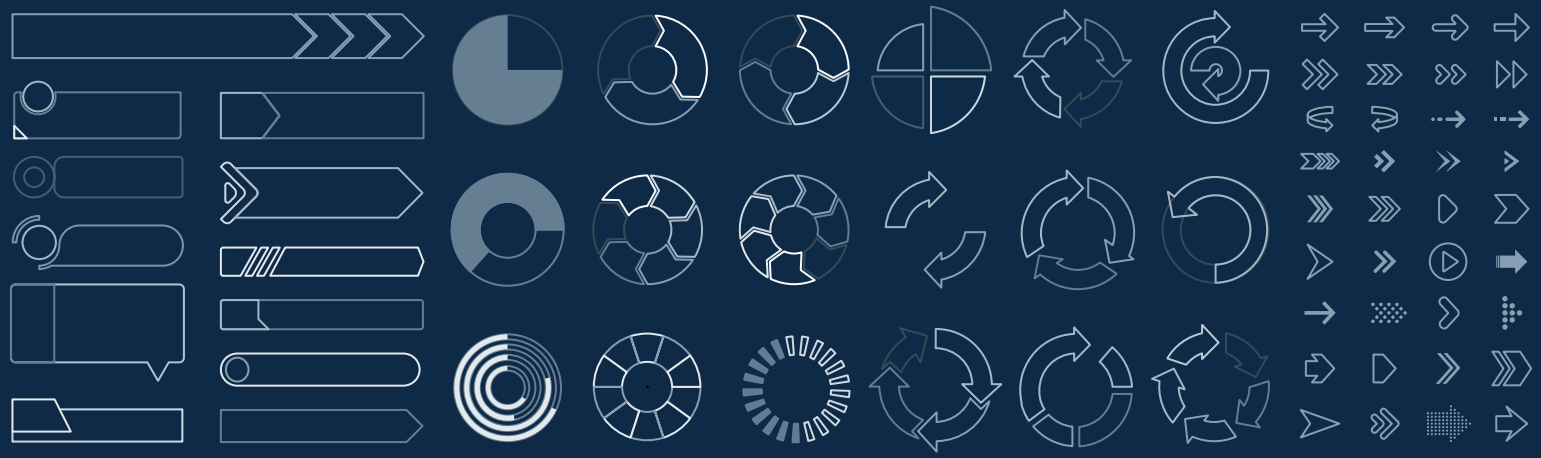
Strategic Test Design

Focusing on test strategy rather than implementation details.

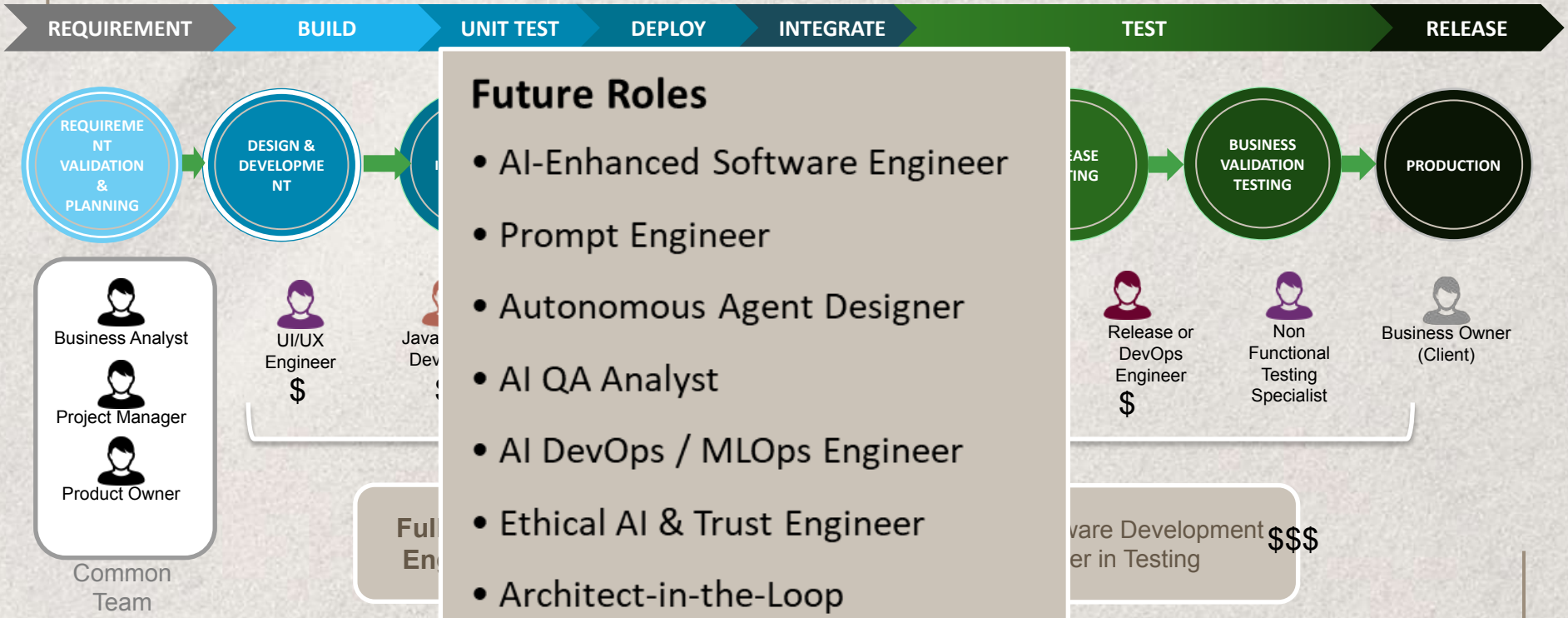
Ethical Oversight

Ensuring AI testing follows quality standards and ethical guidelines.

Evolution of Software Jobs



Emerging Job Roles in IT Industry with AI



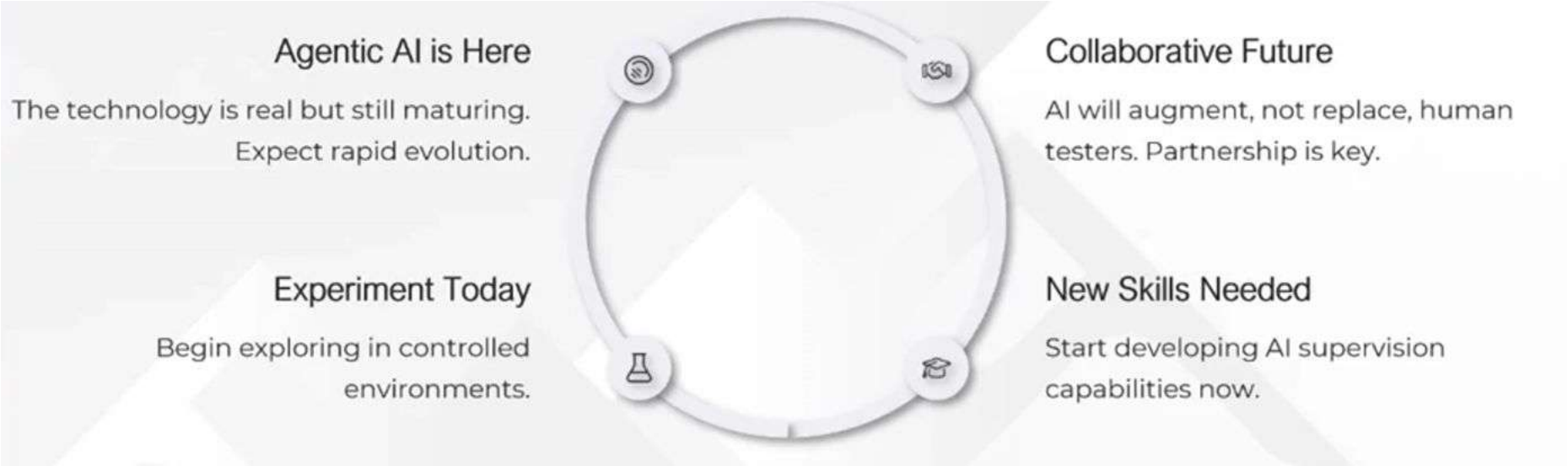
Future Roles

- AI-Enhanced Software Engineer
- Prompt Engineer
- Autonomous Agent Designer
- AI QA Analyst
- AI DevOps / MLOps Engineer
- Ethical AI & Trust Engineer
- Architect-in-the-Loop
- Codebase Coach

Key Characteristics of Future QA Roles

Attribute	Shift
Focus	From writing code → curating, guiding, and integrating AI-generated code
Tools	IDEs + LLMs, Agent Platforms, Prompt Chaining
Skills	Prompt design, critical review, agent orchestration
Collaboration	Human + AI pair programming, cross-agent workflows

Key Takeaways



- ***Less manual, more strategic***
- ***Less mechanical, more creative/abstract***
- ***Less coding line-by-line, more validating and designing systems***
- ***More focused on ethics, governance, and system-level thinking***

Questions ?

Thanks !

kapilalk@gmail.com
+94 71 664 3373
https://www.linkedin.com/in/kapila-karunaratne-a3783926